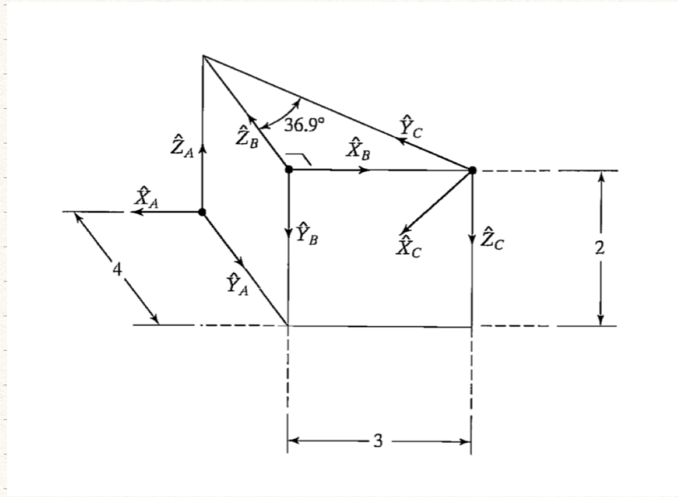
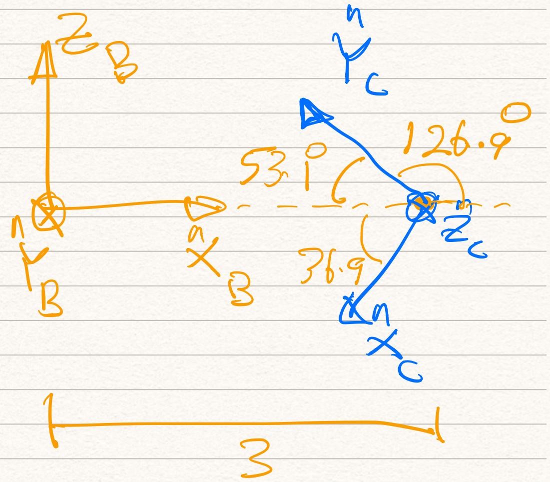
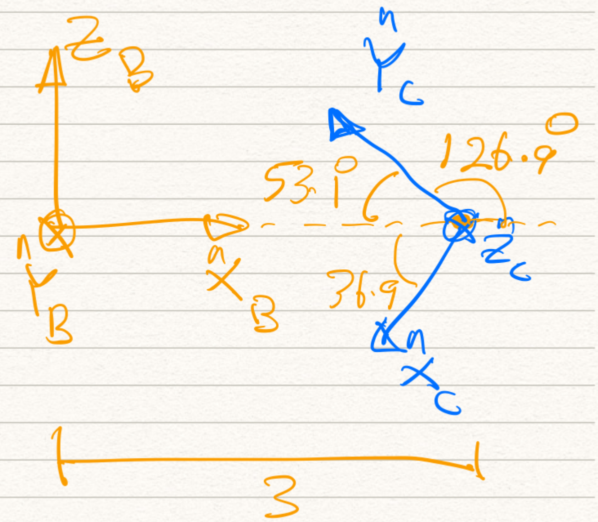


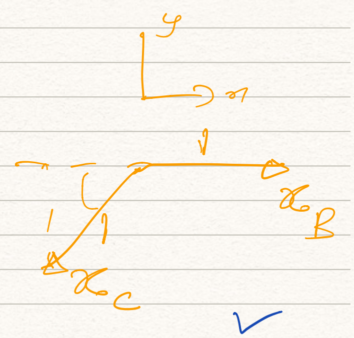


$$
{}^{B}R_{C} = \begin{array}{c} \\ X_B \\ Y_B \\ Z_B \end{array}
\begin{array}{c}
\begin{array}{ccc} X_C & Y_C & Z_C \end{array} \\
\begin{bmatrix} X_C \cdot X_B & Y_C \cdot X_B & Z_C \cdot X_B \\ X_C \cdot Y_B & Y_C \cdot Y_B & Z_C \cdot Y_B \\ X_C \cdot Z_B & Y_C \cdot Z_B & Z_C \cdot Z_B \end{bmatrix}
\end{array}
$$

$$
{}^{B}P_{C} = \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix} \quad \text{— (2)}
$$

$$
{}^{B}T_{C} = \begin{bmatrix} {}^{B}R_{C} & {}^{B}P_{C} \\ 0_{3\times 3} & 1 \end{bmatrix} \quad \text{— (3)}
$$





$$
{}^{B}R_{C} = \begin{bmatrix} -0.7997 & -0.6004 & 0 \\ 0 & 0 & 1 \\ -0.6004 & 0.6004 & 0 \end{bmatrix}
$$

Example:

$x_C \cdot x_B = |x_C|\,|x_B|$

or: $\cos(180+36.9°)$

$= 1 \times 1 \times (-0.7997)$

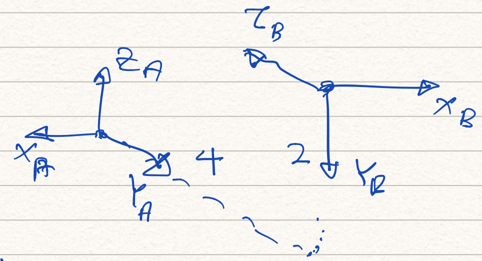
$= -0.7997$

Therefore

$$
{}^{B}T_{C} = \begin{array}{c} \\ x_B \\ Y_B \\ z_B \end{array} \begin{array}{c} \begin{array}{ccc} x_c & Y_c & z_c \end{array} \\ \underbrace{\begin{bmatrix} -0.8 & -0.6 & 0 \\ 0 & 0 & 1 \\ -0.6 & 0.6 & 0 \end{bmatrix}}_{{}^{B}R_{C}} \end{array} \underbrace{\begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix}}_{{}^{B}P_{C}}
$$

$$
{}^A R_B = \begin{bmatrix} x_B \cdot x_A & y_B \cdot x_A & z_B \cdot x_A \\ x_B \cdot y_A & y_B \cdot y_A & z_B \cdot y_A \\ x_B \cdot z_A & y_B \cdot z_A & z_B \cdot z_A \end{bmatrix}
$$

$$
{}^A P_B = \begin{bmatrix} 0 \\ 4 \\ 2 \end{bmatrix}
$$



$$
{}^A R_B = \begin{array}{c} \\ x_A \\ y_A \\ z_A \end{array} \begin{array}{c} \begin{array}{ccc} x_B & y_B & z_B \end{array} \\ \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix} \end{array}
$$

$$
{}^A T_B = \left[\begin{array}{cc} \overbrace{\begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix}}^{{}^A R_B} & \overbrace{\begin{bmatrix} 0 \\ 4 \\ 2 \end{bmatrix}}^{{}^A P_B} \\ \begin{matrix} 0 & 0 & 0 \end{matrix} & 1 \end{array} \right]
$$

Therefore

$${}^{A}T_{C} = {}^{A}T_{B} \times {}^{B}T_{C}$$

$$= \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 \\ 0 & -1 & 0 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -0.8 & -0.6 & 0 & 3 \\ 0 & 0 & 1 & 0 \\ -0.6 & 0.6 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{array}{c} \begin{array}{cc} \quad {}^{A}R_{C} \quad\quad\quad & {}^{A}P_{C} \end{array} \\ \left[\begin{array}{ccc|c} 0.8 & 0.6 & 0 & -3 \\ 0.6 & -0.6 & 0 & 4 \\ 0 & 0 & -1 & 2 \\ 0 & 0 & 0 & 1 \end{array}\right] \end{array}$$

Question-2

- Choice-1 is correct

Question-3

- d is correct

Question-4

- a - Correct

Question-5

a - is correct

Question - 6

b - is correct

Question - 7

d - is correct

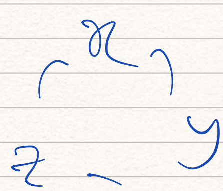
~~Question - 8~~

~~b - is correct~~

Question - 8

b - is correct

Question - 10



x - y - z

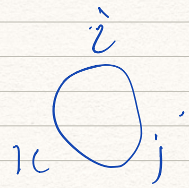
y - z - x

z - x - y

y - x - z

$x - z - y$

$z - y - x$



$$\left(c\tfrac{\theta}{2} + k\, s\tfrac{\theta}{2}\right)\left(ix + jy\right)\left(c\tfrac{\theta}{2} - k\, s\tfrac{\theta}{2}\right)$$

$$= \left[ix\, c\tfrac{\theta}{2} + jy\, c\tfrac{\theta}{2} + jx\, s\tfrac{\theta}{2} - iy\, s\tfrac{\theta}{2}\right]\left(c\tfrac{\theta}{2} - k\, s\tfrac{\theta}{2}\right)$$

$$= \left[i\left(x\, c\tfrac{\theta}{2} - y\, s\tfrac{\theta}{2}\right) + j\left(x\, s\tfrac{\theta}{2} + y\, c\tfrac{\theta}{2}\right)\right]\left(c\tfrac{\theta}{2} - k\, s\tfrac{\theta}{2}\right)$$

$$= i\left(x\, c^2\tfrac{\theta}{2} - y\, s\tfrac{\theta}{2}\cdot c\tfrac{\theta}{2}\right) + j\left(x\, s\tfrac{\theta}{2}\cdot c\tfrac{\theta}{2} + y\, c^2\tfrac{\theta}{2}\right)$$

$$+ j\left(x\, c\tfrac{\theta}{2}\cdot s\tfrac{\theta}{2} - y\, s^2\tfrac{\theta}{2}\right) - i\left(x\, s^2\tfrac{\theta}{2} + y\, c\tfrac{\theta}{2}\cdot s\tfrac{\theta}{2}\right)$$

$$= i\left(x\left(c^2\tfrac{\theta}{2} - s^2\tfrac{\theta}{2}\right) - 2y\, s\tfrac{\theta}{2}\cdot c\tfrac{\theta}{2}\right)$$

$$+ j\left(2x\, s\tfrac{\theta}{2}\cdot c\tfrac{\theta}{2} + y\left(c^2\tfrac{\theta}{2} - s^2\tfrac{\theta}{2}\right)\right)$$

$$= i\left(x\cos\theta - y\sin\theta\right) + j\left(x\sin\theta + y\cos\theta\right)$$

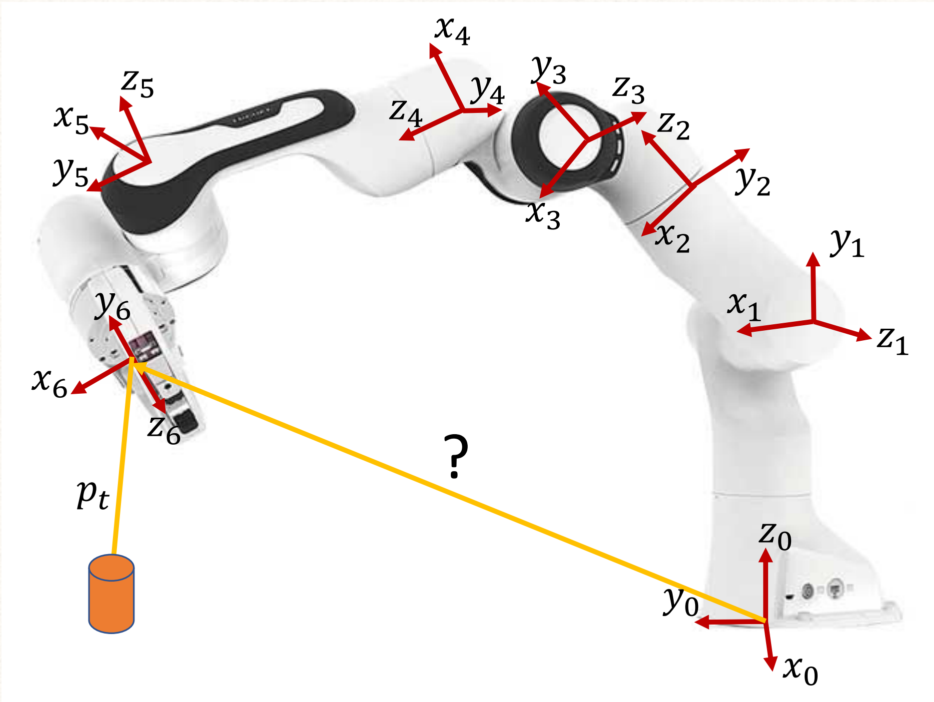
$$= \underbrace{\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}}_{R.} \begin{pmatrix} x \\ y \end{pmatrix}$$

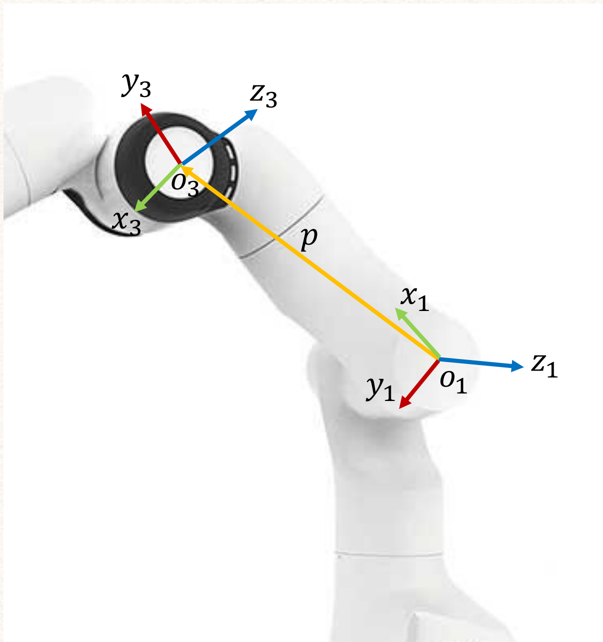
3D Kinematics tutorial-1

Learning outcomes:

- Know what the forward kinematics problem is
- Know how to assign frames to joints in a 3D robot.
- Know what a homogeneous coordinate system is.
- Know how to interpret a homogeneous transformation matrix.

The forward kinematics problem Given knowledge of only the joint angles and link lengths, how to determine the position and orientation of the end-effector of a robot relative the the base.

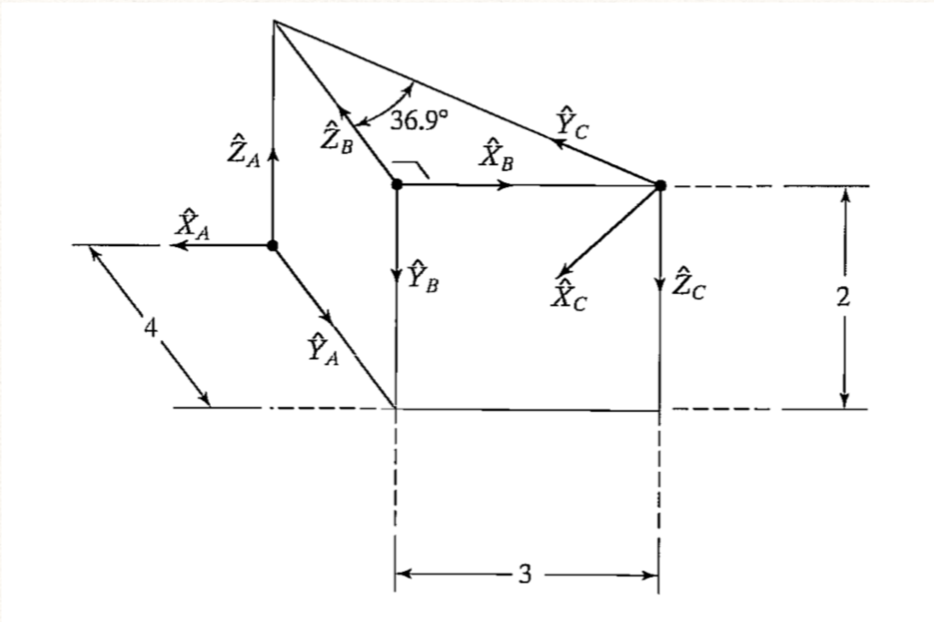




A coordinate frame i consists of an origin, denoted O_i, and three mutually orthogonal basis vectors, denoted (x_i, y_i, z_i), that are fixed within a given body.

Question-1

Craig, Q 2.32 Find the transformation matrices ${}^{B}T_C$, ${}^{A}T_B$, and ${}^{A}T_C$.



Question-2

You found a piece of paper with a set of rotation matrices. However, some of the numbers are not clear. Which of the following rotation matrices could be the complete set?

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ ? & ? & ? \end{bmatrix} \quad \begin{bmatrix} ? & 0 & 0.7071 \\ ? & 1 & 0 \\ ? & 0 & 0.7071 \end{bmatrix} \quad \begin{bmatrix} 0.866 & ? & 0 \\ 0.5 & ? & 0 \\ 0 & ? & 1 \end{bmatrix}$$

a)
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \quad \begin{bmatrix} 0.7071 & 0 & 0.7071 \\ 0 & 1 & 0 \\ -0.7071 & 0 & 0.7071 \end{bmatrix} \quad \begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

b)
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix} \quad \begin{bmatrix} 0.7071 & 0 & 0.7071 \\ 0 & 1 & 0 \\ 0 & 0 & 0.7071 \end{bmatrix} \quad \begin{bmatrix} 0.866 & 0 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

c)
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix} \quad \begin{bmatrix} -0.7071 & 0 & 0.7071 \\ 0 & 1 & 0 \\ -0.7071 & 0 & 0.7071 \end{bmatrix} \quad \begin{bmatrix} 0.866 & 0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

d)
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \quad \begin{bmatrix} -0.7071 & 0 & 0.7071 \\ 0 & 1 & 0 \\ 1 & 0 & 0.7071 \end{bmatrix} \quad \begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.5 & -0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Question-3

Which of the following give the correct expressions for rotation angles given the following rotation matrix?

$$
\begin{aligned}
{}^{i}\boldsymbol{R}_j(\alpha,\beta,\gamma) &= \begin{bmatrix} c\alpha c\beta c\gamma - s\alpha s\gamma & -c\alpha c\beta s\gamma - s\alpha c\gamma & c\alpha s\beta \\ s\alpha c\beta c\gamma - c\alpha s\gamma & -s\alpha c\beta s\gamma + c\alpha c\gamma & s\alpha s\beta \\ -s\beta c\gamma & s\beta s\gamma & c\beta \end{bmatrix} \\
&= \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}
\end{aligned} \tag{1}
$$

a)

$$\beta = atan2(\sqrt{r_{33}^2 + r_{32}^2}, r_{31}); \alpha = atan2(\frac{r_{32}}{s\beta}, -\frac{r_{31}}{s\beta}); \gamma = atan2(\frac{r_{23}}{s\beta}, \frac{r_{13}}{s\beta}) \tag{2}$$

b)

$$\beta = atan2(r_{31}, \sqrt{r_{32}^2 + r_{33}^2}); \alpha = atan2(\frac{r_{21}}{c\beta}, \frac{r_{11}}{c\beta}); \gamma = atan2(\frac{r_{32}}{c\beta}, \frac{r_{33}}{c\beta}) \tag{3}$$

c)

$$\beta = atan2(r_{31}, \sqrt{r_{32}^2 + r_{33}^2}); \alpha = atan2(\frac{r_{23}}{s\beta}, \frac{r_{13}}{s\beta}); \gamma = atan2(\frac{r_{32}}{s\beta}, -\frac{r_{31}}{s\beta}) \tag{4}$$

d)

$$\beta = atan2(\sqrt{r_{31}^2 + r_{32}^2}, r_{33}); \alpha = atan2(\frac{r_{23}}{s\beta}, \frac{r_{13}}{s\beta}); \gamma = atan2(\frac{r_{32}}{s\beta}, -\frac{r_{31}}{s\beta}) \tag{5}$$

Question-4

"A rotation of a frame in 3D space around an axis passing through its origin can be realized through 3 successive rotations around any 3 linearly independent axes". Is this statement correct?

a) Correct

b) Not correct

Question-5

The homogeneous transformation matrix from the end point frame to the base frame of a 3-link manipulator is given by

$$
{}^{0}\boldsymbol{T}_4 = \begin{bmatrix} \cos(\theta_1+\theta_2+\theta_3) & -\sin(\theta_1+\theta_2+\theta_3) & 0 & L_1\cos(\theta_1) + L_2\cos(\theta_1+\theta_2) \\ \sin(\theta_1+\theta_2+\theta_3) & \cos(\theta_1+\theta_2+\theta_3) & 0 & L_1\sin(\theta_1) + L_2\sin(\theta_1+\theta_2) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{6}
$$

The Jacobian matrix of the above manipulator is given by

a)

$$\boldsymbol{J} = \begin{bmatrix} -L_2\sin(\theta_1+\theta_2) - L_1\sin(\theta_1) & -L_2\sin(\theta_1+\theta_2) & 0 \\ L_2\cos(\theta_1+\theta_2) + L_1\cos(\theta_1) & L_2\cos(\theta_1+\theta_2) & 0 \\ 0 & 0 & 0 \end{bmatrix} \tag{7}$$

b)

$$\boldsymbol{J} = \begin{bmatrix} -L_2\sin(\theta_1+\theta_2) - L_1\sin(\theta_1) & 0 & -L_2\sin(\theta_1+\theta_2) \\ L_2\cos(\theta_1+\theta_2) + L_1\cos(\theta_1) & L_2\cos(\theta_1+\theta_2) & 0 \\ 0 & 0 & 0 \end{bmatrix} \tag{8}$$

c)

$$\boldsymbol{J} = \begin{bmatrix} -L_2\sin(\theta_1+\theta_2) - L_1\sin(\theta_1) & L_2\cos(\theta_1+\theta_2) + L_1\cos(\theta_1) & 0 \\ -L_2\sin(\theta_1+\theta_2) & L_2\cos(\theta_1+\theta_2) & 0 \\ 0 & 0 & 0 \end{bmatrix} \tag{9}$$

Question-6

What would be the joint torques needed to exert a force vector $\boldsymbol{f} = [1\ 1\ 1]^T$ at the tip of the 3-link manipulator with the homogeneous transformation matrix:

$${}^{0}\boldsymbol{T}_4 = \begin{bmatrix} \cos(\theta_1+\theta_2+\theta_3) & -\sin(\theta_1+\theta_2+\theta_3) & 0 & L_1\cos(\theta_1) + L_2\cos(\theta_1+\theta_2) \\ \sin(\theta_1+\theta_2+\theta_3) & \cos(\theta_1+\theta_2+\theta_3) & 0 & L_1\sin(\theta_1) + L_2\sin(\theta_1+\theta_2) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{10}$$

a)

$$\boldsymbol{\tau} = \begin{bmatrix} L_2\sin(\theta_1+\theta_2) + L_1\sin(\theta_1) + L_2\cos(\theta_1+\theta_2) + L_1\cos(\theta_1) \\ -L_2\sin(\theta_1+\theta_2) + L_2\cos(\theta_1+\theta_2) \\ 0 \end{bmatrix} \tag{11}$$

b)

$$\boldsymbol{\tau} = \begin{bmatrix} -L_2\sin(\theta_1+\theta_2) - L_1\sin(\theta_1) + L_2\cos(\theta_1+\theta_2) + L_1\cos(\theta_1) \\ -L_2\sin(\theta_1+\theta_2) + L_2\cos(\theta_1+\theta_2) \\ 0 \end{bmatrix} \tag{12}$$

c)

$$\boldsymbol{\tau} = \begin{bmatrix} -L_2\sin(\theta_1+\theta_2) - L_1\sin(\theta_1) - L_2\sin(\theta_1+\theta_2) \\ L_2\cos(\theta_1+\theta_2) + L_1\cos(\theta_1) + L_2\cos(\theta_1+\theta_2) \\ 0 \end{bmatrix} \tag{13}$$

Question-7

What condition in joint angles in the 3-link manipulator with the homogeneous transformation matrix given by

$$^{0}\boldsymbol{T}_4 = \begin{bmatrix} \cos(\theta_1+\theta_2+\theta_3) & -\sin(\theta_1+\theta_2+\theta_3) & 0 & L_1\cos(\theta_1)+L_2\cos(\theta_1+\theta_2) \\ \sin(\theta_1+\theta_2+\theta_3) & \cos(\theta_1+\theta_2+\theta_3) & 0 & L_1\sin(\theta_1)+L_2\sin(\theta_1+\theta_2) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{14}$$

will result in a torque only around joint-1 to exert a force vector $\boldsymbol{f} = [1\ 1\ 1]^T$ at the tip?

a) $\theta_2 = 0$

b) $\theta_2 + \theta_3 = \pi/2$

c) $\theta_2 + \theta_3 = \pi/4$

d) $\theta_1 + \theta_2 = \pi/4$

Question-8

Which of the following vectors is in a homogeneous coordinate notation?

a)

$$\boldsymbol{v}_1 = \begin{bmatrix} 2 & 4 & 1 & 0 \end{bmatrix}^T \tag{15}$$

b)

$$\boldsymbol{v}_2 = \begin{bmatrix} 4 & 8 & 2 & 2 \end{bmatrix}^T \tag{16}$$

c)

$$\boldsymbol{v}_3 = \begin{bmatrix} 4i & 8j & 2k \end{bmatrix}^T \tag{17}$$

Question-9

a) What is the purpose of the first rotation of a $Z-Y-X$ Euler rotation?
b) Likewise, what is the purpose of the second rotation?

Question-10

How many sequences are there for Euler rotations? i.e. $Z-Y-X$, $X-Y-Z$, ...
a) For moving frame rotations b) For fixed frame rotations